$$\frac{}{\Gamma \vdash \mathsf{get}^{\mathcal{S}}() : \tau \rightarrowtail \mathtt{perform}\ \mathcal{G}\ \tau\ \mathsf{unit}}\ \text{Compile-Get-State}$$

$$\frac{\Gamma \vdash e : \tau \rightarrowtail e_1}{\Gamma \vdash \mathsf{modify}^{\mathcal{S}}(\delta, e) : \mathsf{unit} \rightarrowtail \mathtt{perform}\ \mathcal{M}\ \tau\ e_1}\ \text{Compile-Modify-State}$$

$$\mathtt{h}_{\mathcal{C}} : \mathtt{lab} \quad = \quad \{\mathcal{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\}$$

$$\mathtt{h}_{\mathcal{S}} : \mathtt{lab} \quad = \quad \{\mathcal{G} \to \forall\alpha.\ \mathsf{unit} \to \alpha; \mathcal{M} \to \forall\alpha.\alpha \to \mathsf{unit}\}$$

$$\mathsf{wrap}_{\mathcal{C}} \quad = \quad \mu w : \big(() \to^{\langle \mathtt{h}_{\mathcal{C}}\rangle} \mathsf{bool}\big) \to \mathsf{bool}.\ \lambda f : \big(() \to^{\langle \mathtt{h}_{\mathcal{C}}\rangle} \mathsf{bool}\big).$$

$$\mathtt{handler}\ \{\mathcal{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if}\ w\ (\lambda_.e\ v)\ \text{then}\ k\ v\ \text{else}\ \mathtt{error}_l\}\ f$$

where $\lambda\langle l, \langle e, v\rangle\rangle$ deconstructs the argument as tuple pattern

$$\mathsf{wrap}_{\mathcal{S}} \quad = \quad \Lambda\alpha.\lambda f : \mathsf{unit} \to \alpha.$$

$$\mathtt{handler}\ \{\mathcal{G} \to \lambda x.\lambda k.\lambda s.k\ s\ s; \mathcal{M} \to \lambda x.\lambda k.\lambda s.k\ s\ x;\}\ f_1$$

where $f_1 = \lambda_.((\lambda x.\lambda s.\langle x, s\rangle)\ (f\ \mathsf{unit}))$

$$\mathsf{wrap} \quad = \quad \forall\alpha.\big(\big(() \to^{\langle \mathtt{h}_{\mathcal{C}}\rangle} \alpha\big) \to \alpha\big).\ \lambda f : \big(() \to^{\langle \mathtt{h}_{\mathcal{C}}\rangle} \alpha\big).$$

$$\mathtt{handler}\ \{\mathcal{C} \to$$

$$\lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\lambda s.(\lambda\langle b, s_1\rangle.\ \text{if}\ b\ \text{then}\ k\ v\ s_1\ \text{else}\ \mathtt{error}_l)\ e_1\}\ f_1$$

where $f_1 = \lambda_.((\lambda x.\lambda s.x)\ (f\ \mathsf{unit}))$

where $e_1 = \mathsf{wrap}_{\mathcal{S}}\ \lambda_.(\mathsf{wrap}_{\mathcal{C}}\ (\lambda_.e\ v))\ s$

where $\lambda\langle l, \langle e, v\rangle\rangle$ and $\lambda\langle b, s_1\rangle$ deconstructs the argument as tuple pattern

Figure 1: Compiler Rules for Contract State Handling